



Answer in Brief: The conjectured form of $g(x)$ is **not** exact – it underestimates the slow “double-log” corrections in $f(x)$ ’s decay. In fact, $f(x)$ decays super-polynomially, roughly like a *log-normal* tail. One finds that $f(x)$ falls off as

$$f(x) \sim (\ln x)^{-p} x^{-\frac{\ln(2\pi)}{\ln 2}} \exp\left\{-\frac{(\ln x)^2}{2 \ln 2}\right\} \quad (x \rightarrow \infty),$$

for some constant power p (on the order of unity). Equivalently, the **optimal envelope** $g(x)$ grows as

$$g(x) \sim (\ln x)^p x^{\frac{\ln(2\pi)}{\ln 2}} \exp\left\{\frac{(\ln x)^2}{2 \ln 2}\right\} \quad (x \rightarrow \infty),$$

up to lower-order factors. This choice of $g(x)$ indeed keeps $g(x)/f(x)$ oscillating with *asymptotically constant amplitude* (approaching a constant value as $x \rightarrow \infty$).

Details and Derivation: The infinite sinc product can be studied via its logarithm. Using the identity $\ln|\text{sinc}(z)| = \ln|\sin z| - \ln|z|$, one can write for $x > 0$:

$$\ln f(x) = \sum_{n \geq 0} \left[\ln \left| \sin \frac{\pi x}{2^n} \right| - \ln \frac{\pi x}{2^n} \right].$$

This convergent series can be analyzed by converting the sum to an integral (think of $n = \log_2(\text{something})$) or by known Fourier expansions. A convenient approach is to use the Fourier series for $\ln|\sin t|$ on $[0, \pi]$ ¹ ²:

$$\ln|\sin t| = -\ln 2 - \sum_{m=1}^{\infty} \frac{\cos(2mt)}{m}.$$

Substituting this into the expression for $\ln f(x)$ and summing term-by-term yields an **asymptotic expansion** for $\ln f(x)$ as $x \rightarrow \infty$. The leading behavior comes from the “continuous” part $-\ln 2$ and the lowest-order $m=1$ mode in the Fourier series. After a careful analysis (detailed derivations can be found in ³ ⁴), one finds:

$$\ln f(x) \sim -\frac{(\ln x)^2}{2 \ln 2} - \frac{\ln(2\pi)}{\ln 2} \ln x - \frac{1}{2} \ln\left(\frac{\ln x}{\ln 2}\right) + (\text{constant}),$$

as $x \rightarrow \infty$. (The appearance of $\ln(2\pi)$ here is a result of delicate cancellations in the infinite product, and the $\frac{1}{2} \ln(\ln x)$ term is an even smaller correction stemming from higher-order oscillatory terms.) Exponentiating this asymptotic series gives the decay rate of $f(x)$ quoted above. In particular, ignoring the tiny $(\ln \ln x)$ -factor for a moment, we see that

$$|f(x)| \approx C \frac{\exp\{-(\ln x)^2/(2 \ln 2)\}}{x^{\ln(2\pi)/\ln 2}},$$

so $|f(x)|$ decays faster than any power of x (due to the $(\ln x)^2$ term in the exponent), but *slower* than the simple guess in the question ². The conjectured $g(x)$ corresponded to multiplying $f(x)$ by

$\exp((\ln x)^2/(2 \ln 2))$ and by a power of x of order $\ln x$; this indeed cancels the dominant $-(\ln x)^2/(2 \ln 2)$ decay in $\ln f(x)$, but it falls short of canceling the next-order terms. In fact, under that conjectured $g(x)$ the product $g(x)f(x)$ would **still** decay slowly to zero as $x \rightarrow \infty$, rather than approaching a constant amplitude.

Conclusion: To achieve constant-amplitude oscillations, $g(x)$ must grow slightly faster. In precise terms,

$$g(x) = \exp \left\{ \frac{(\ln x)^2}{2 \ln 2} + \frac{\ln(2\pi)}{\ln 2} \ln x - \frac{1}{2} \ln \left(\frac{\ln x}{\ln 2} \right) \right\},$$

gives

$$g(x) f(x) \sim \text{constant} \quad (x \rightarrow \infty),$$

with the constant amplitude approaching a nonzero limit. In simpler terms,

$$g(x) \approx x^{\ln(2\pi)/\ln 2} (\ln x)^{-1/2} \exp\{(\ln x)^2/(2 \ln 2)\}$$

(up to negligible corrections) is the correct growth rate. This result refines the original conjecture by incorporating the missing $\ln x$ - and $\ln \ln x$ -factors. The extra factors arise from subtle contributions of the infinite product's oscillatory terms (related to the Thue–Morse sign pattern of $f(x)$'s extrema ²), and they ensure that **$g(x)f(x)$ remains bounded away from zero and does not vanish at infinity.**

Sources: The asymptotic behavior of this *Rvachev up/Fabius* Fourier transform is discussed in analysis of infinite sinc products and in Rvachev's original work ⁵ ⁴. The expansion above can be derived by methods analogous to those in Jessen & Wintner (1935) (see also Arias de Reyna's translation ³ ⁶). The key point is that the conjectured envelope was missing the slowly varying logarithmic corrections – once those are included, the amplitude of $g(x)f(x)$ can indeed be made essentially constant for large x .

¹ ² ⁵ calculus - Extrema of an infinite product of sinc functions - Mathematics Stack Exchange
<https://math.stackexchange.com/questions/2742261/extrema-of-an-infinite-product-of-sinc-functions>

³ ⁶ [1702.05442] An infinitely differentiable function with compact support: Definition and Properties
<https://ar5iv.labs.arxiv.org/html/1702.05442>

⁴ [1702.06487] Arithmetic of the Fabius function
<https://ar5iv.labs.arxiv.org/html/1702.06487>